

INFORMATICS INSTITUTE OF TECHNOLOGY

CW1 Individual Contribution Report

3D Interactive Media Product

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Module code - 5MMCS007W

Group and role - Group 03 / Student 1

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Introduction

For Coursework 1, our Group 03 worked on creating an interactive LEGO island using Unity. My role as Student 1 was to build the beach starting area. I added the sandy shoreline, palm trees, rocks and a marker to show where the player would start. I also kept some space on the sand to test the red LEGO brick and placed a rock nearby to check its collision.

I worked on making the LEGO brick move using the WASD keys by applying force in different directions. I also added a button to play the LEGO minifigure's walking animation. While testing, I checked whether the brick could move on the sand and stop when it touched the rock. I also clicked the animation button several times to check that the minifigure stayed in the camera view and that the walking animation could play again. In this report, I explain how I created the beach scene, the problems I faced and how I fixed them, and how I prepared my work to share with the group.

Beach Starting Area

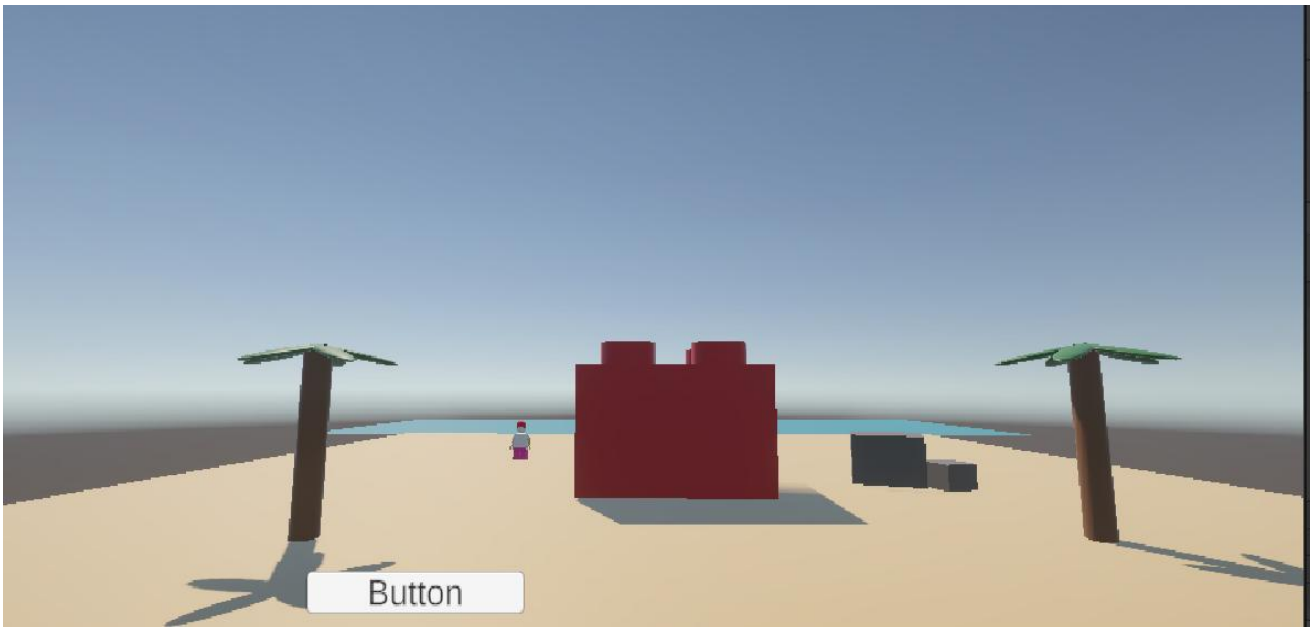


Figure 1

I used a plane as the base of the beach. Applied a sand-coloured material to it. I made two rocks of sizes and put them together next to the path of the red LEGO brick. The rocks added detail to the beach. Gave the beach an obstacle for testing the brick's movement. I also built a palm tree using a cylinder for the trunk and flattened spheres for the leaves. After arranging the leaves, around the top of the trunk I grouped the pieces. Duplicated the palm tree. I placed one palm tree on either side of the beach to leave space between them.

I added an empty GameObject named PlayerStart to mark the player's starting position on the sand. I placed it away from the test path made of bricks. Then I moved the minifigure close to that marker. I grouped the ground, rocks, palms, the starting point and the water under a folder named BeachStartArea in the Hierarchy. This helped me quickly find all the objects that were part of my area when getting the scene ready, for the group.

The sand looked like an open area before I added the water. I put a plane next to it and gave the plane a blue material. During setup the water went far across the sand and the minifigure seemed to be standing in it. I changed to a down view to check where the two surfaces overlapped. I then fixed the water planes position and size until it touched the edge of the sand. Figure 1 shows the shoreline, the two palms and the rock cluster, in the beach layout.

Rigidbody Brick and WASD Controls

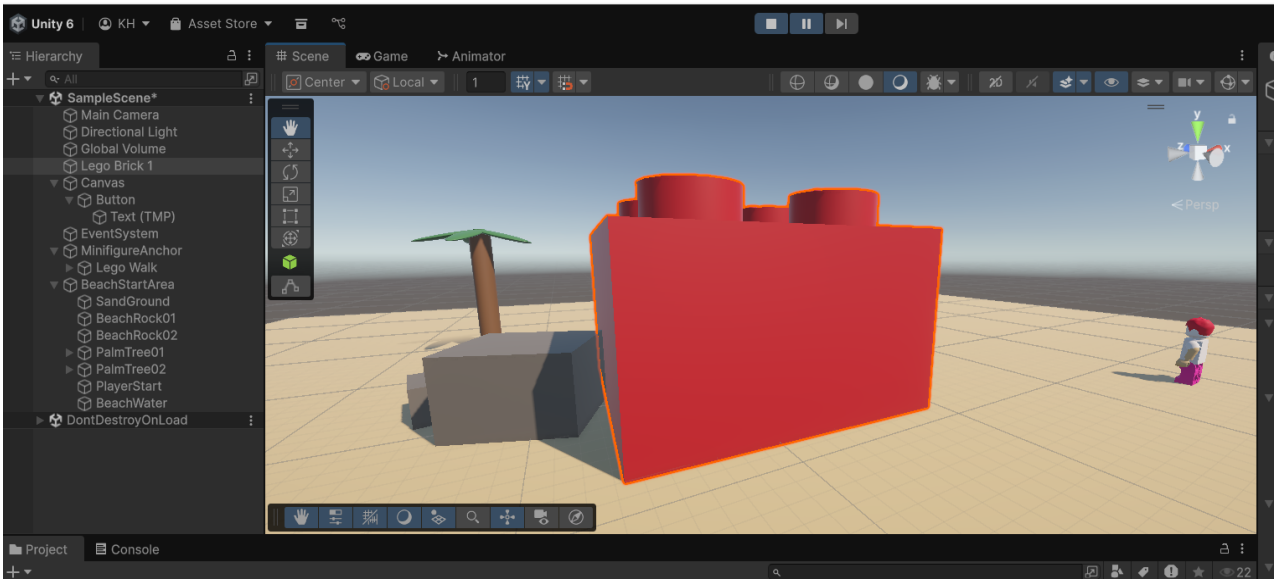


Figure 2

I imported the LEGO brick model from Tutorial 02–03 and placed it in my Unity scene. After that, I added a Rigidbody component and enabled gravity so the brick could fall. I also added a Box Collider and adjusted its center and size to fit the brick. The ground already had a collider, so I didn't need to add another one. To check if everything worked, I moved the brick above the ground and entered Play mode. The brick fell onto the ground and stopped when it touched the surface.

I used the PhysicsController.cs file given by our lecturer to make the LEGO brick move using the keyboard. The script gets the Rigidbody component when the game starts. In FixedUpdate(), it checks which keys I am pressing and applies force to move the brick. The A and D keys move the brick along the X axis, while W and S move it along the Z axis. I could also change the speed value in the Inspector to control how much force was applied. I tried changing this value in Unity to see how it affected the brick's movement.

When I tested the brick with Speed set to 10, it responded to all four movement keys while falling, but after landing on the sand it barely moved. This showed me that the keyboard controls were working, but the speed was too low to move the brick properly on the ground. I changed the Speed value to 30 and tested it again after the brick landed. This time the brick moved across the sand. I also tested what happened when the brick hit a rock. I held the A key until the brick reached the rock, and it stopped moving when it touched the rock even though was still pressing the key. I paused the game in Play mode and took a screenshot of this result for Figure 2.

Minifigure Animation and Button

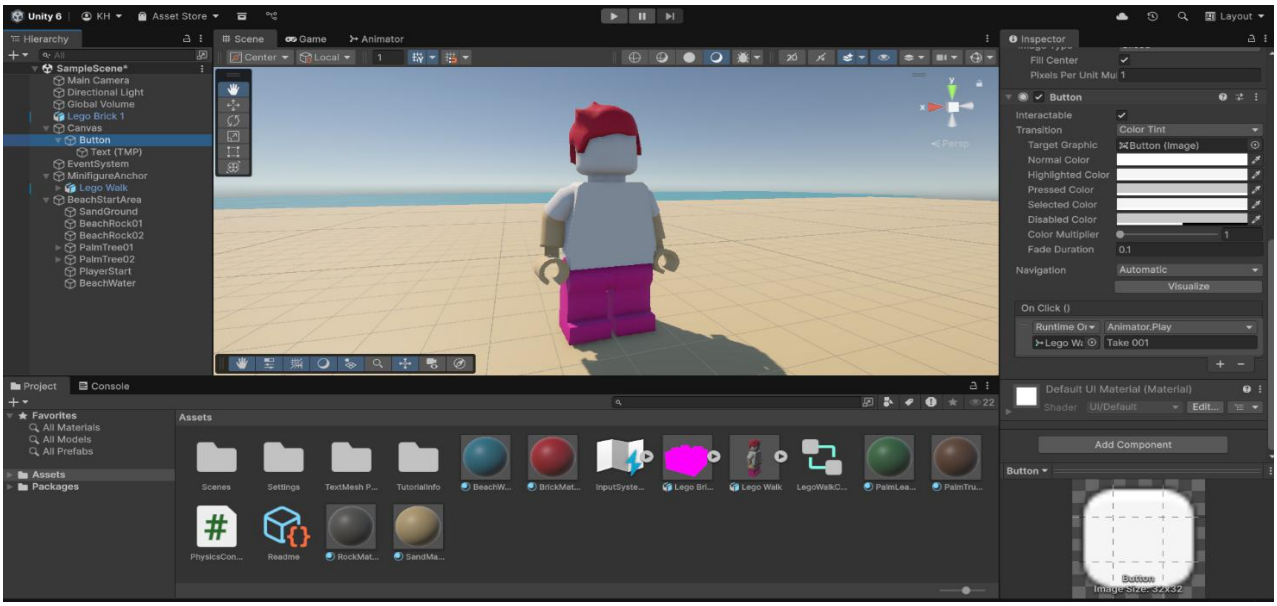


Figure 3

I imported the Lego Walk.FBX model given by our lecturer and checked the Take 001 animation in Unity. The animation showed the LEGO minifigure walking in the same place without moving forward. After that, I created an Animator Controller and connected it to the minifigure's Animator component. In the Animator window, I added Take 001 for the walking animation and also created an Idle state. I set Idle as the default state because I wanted the minifigure to stay still when I started the game.

After that, I added a UI button inside a Canvas in Unity. I went to the button's On Click settings and selected the minifigure's Animator. Then I chose Animator.Play and entered Take 001 as the animation state name. Figure 3 shows the minifigure and the button settings I used. This allowed me to test the walking animation by pressing the button in the Game view.

When I clicked the button for the first time, the walking animation worked, but the minifigure disappeared from the camera view. I checked its Transform before and after pressing the button and noticed that the animation was changing its position. To fix this, I created an empty parent object called MinifigureAnchor and placed the minifigure inside it. This helped me control where the minifigure was placed while the walking animation played. I adjusted the anchor's position to keep the minifigure near the starting point and tested the animation again to make sure it stayed visible.

I also noticed that the minifigure stopped after the first walking animation, and when I clicked the button again, nothing happened. To fix this, I added a transition from Take 001 back to Idle in the Animator. I set Exit Time to 1 and Transition Duration to 0. After making these changes, I tested the button again. I clicked it once, waited for the walking animation to finish, and clicked it again. This time the animation worked on both clicks.

Final Testing and Group Handoff

After finishing the beach scene, I tested the LEGO brick and minifigure again. I moved the brick a little lower so it started just above the sand and checked if the WASD controls were still working. I also changed the camera position so I could see the beach, palm trees, rocks and both LEGO objects in the Game view. After that, I took a wide screenshot of the beach scene and another closer screenshot of the minifigure to show the animation setup more clearly.

After finishing my work, I saved the scene as Student1_Beach and exported it as a Unity package. I included the dependencies so the other group members could use the assets needed for the scene. Then I uploaded the Student1_Beach.unitypackage file to our group's GitHub repository so they could add my work to the main group project.

AI Use Reflection

I built the beach area in Unity and added the rocks, palm trees and minifigure animation button. I tested the WASD controls and changed the brick's speed because it was barely moving on the sand. I also checked that the brick stopped when it hit a rock. For the minifigure, I made sure it stayed in the camera view and that the walking animation worked when I clicked the button again. I used ChatGPT to help me understand some Unity settings and fix problems I faced during the project. It also gave me examples of C# code for the WASD controls, which I used as a guide when working on my own code and testing it in Unity. I also used ChatGPT to help me plan the report by asking for an outline and example paragraphs.

AI Chat Summary

ChatGPT helped Student 1 understand some Unity settings and troubleshoot problems during the CW1 coursework. The student built the beach environment by adding rocks, palm trees and the shoreline, and worked on the LEGO brick movement and minifigure animation button. ChatGPT provided guidance when the student faced difficulties with Unity settings and when the minifigure moved out of the camera view.

For the WASD controls, ChatGPT provided example C# code lines that the student used as a guide while working on PhysicsController.cs. The student added and tested the controls in Unity, adjusted the brick's speed from 10 to 30 because it was moving too slowly, and confirmed that the brick stopped when it collided with a rock. The student also tested the minifigure's walking animation to make sure it played again when the button was clicked.

ChatGPT also helped the student organise the individual coursework report by providing an outline and example paragraphs. The student carried out the practical work, made adjustments and tested the results in Unity, while using ChatGPT mainly for guidance, troubleshooting and report planning.